

Cellular Expert

5. CE Express. Importing data

1. Objective

This module explains how to **import external data into CE Express** and how imported data becomes part of the project environment. It covers the import of network objects from CSV files and antenna patterns from external files, as well as the use of mapping templates to ensure consistency and repeatability.

By the end of this exercise, participants will be able to:

- Import network objects using CSV files
- Map external data fields to CE Express object attributes
- Create and reuse import mapping templates
- Import antenna pattern files into the CE Express database
- Verify and review imported data within the workspace

2. Understanding Data Import in CE Express

Data import is a key capability in CE Express, allowing users to bring information from external sources into a workspace. Imported data may originate from:

- External planning tools
- Inventory systems
- Field surveys
- Desktop-based CE projects

CE Express supports structured import workflows to ensure that imported data is:

- Correctly mapped to internal object models
- Consistent across multiple imports
- Easy to review and validate after import

3. Initial Data and Prerequisites

This exercise assumes:

- A prepared workspace created in previous exercises
- Prepared geodata loaded in the workspace
- Access to import files stored locally on the computer

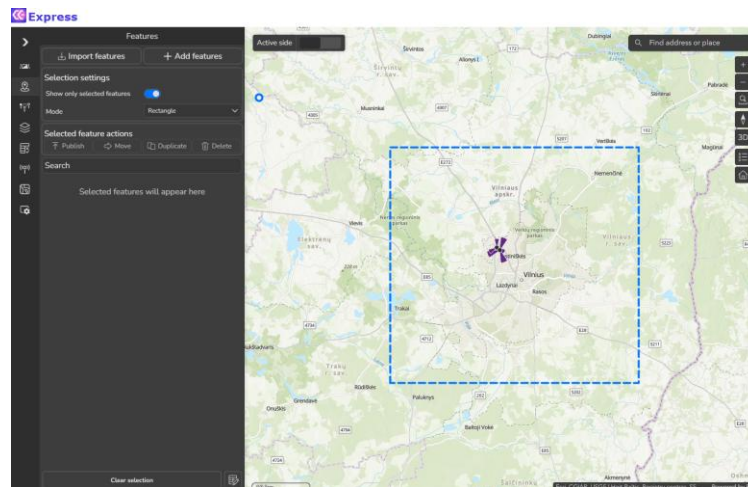
4. Exercise

4.1 Step 1 – Open the Workspace

1. Open the CE Express application:

https://cecom2.cellular-expert.com/ce_express/

2. From the workspace list, select the workspace used in the previous exercise.



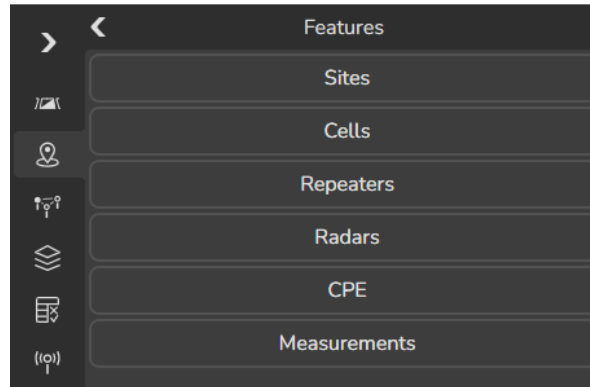
4.2 Step 2 – Importing Network Objects (Cells and Sites)

4.2.1 Purpose of Network Object Import

Importing network objects allows users to rapidly populate a workspace with sites and cells instead of creating them manually. This approach is especially useful when working with existing datasets or migrating data between systems.

4.2.2 Opening the Import Tool

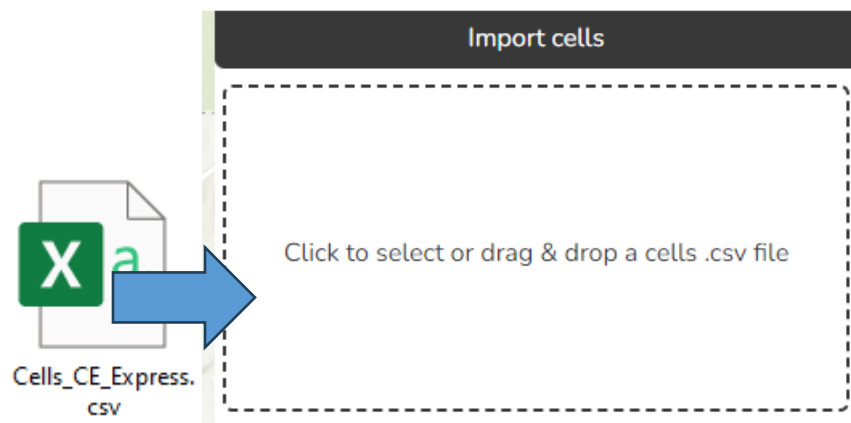
1. Open the **Features** tool.
2. Click **Import Features**.
3. Select **Cells**.



The Import dialog opens on the right side of the screen.

4.2.3 Loading the CSV File

1. Open **Windows File Explorer**.
2. Navigate to:
C:\CE_Course\ImportingData\Network
3. Select the file **Cells_CE_Express.csv**.
4. Drag and drop the file into the Import dialog.



The CSV file is uploaded and ready for mapping.

4.2.4 Defining Import Options

Enable the following options:

- **Create Sites** – automatically creates site objects based on site-related fields

- **Use Mapping** – allows explicit mapping between CSV fields and CE Express attributes

The screenshot shows a dark-themed dialog box titled "Import features". Inside, there's a section labeled "Importing" with two rows: "cells" with a value of "15" and "Sites" with a value of "as per Site name field". Below this section, there are two toggle switches: "Use mapping" and "Create sites", both of which are turned on (blue).

These options ensure that both sites and cells are created correctly during import.

4.2.5 Mapping CSV Fields to CE Express Attributes

Mapping defines how columns in the CSV file are translated into CE Express object parameters.

For each attribute, define the following mappings:

CE Express	CSV file	Fill
Cell name	Cell Name	Leave empty
X	longitude	Leave empty
Y	latitude	Leave empty
Azimuth	Azimuth	Leave empty
Site name	Tower Name	Leave empty
Height	Height Above Ground	Leave empty
Downtilt	Tilt	Leave empty
El. Downtilt	Leave empty	0
Frequency	Frequency	Leave empty
Power	Cell Power	Leave empty
Misc loss, dB	Leave empty	0
Bandwidth	Bandwidth	Leave empty
Noise figure	Leave empty	6
Downlink duplex factor	Leave empty	0.6
Subcarrier spacing	Leave empty	30

TX MIMO	TxMIMO	Leave empty
RX MIMO	RxMIMO	Leave empty
Active antenna effect	Leave empty	6
Cell load, %	Leave empty	30
Color index	Leave empty	Leave empty
Technology	Tech	Leave empty
Prediction model ID	Leave empty	3
Prediction model configuration ID	Leave empty	4
Frequency group	frequency_group	Leave empty
antenna_id	Leave empty	515
carriers	Leave empty	[]
site_id	Leave empty	Leave empty
duplex_mode	Duplex	Leave empty

Required mapping

	Source	Fill
Cell name	Cell Name	
X	longitude	
Y	latitude	
Azimuth	Azimuth	

Optional mapping

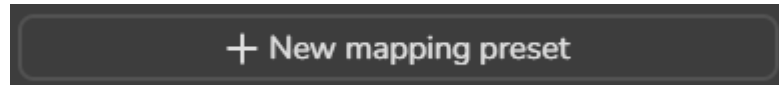
	Source	Fill
Site name	Tower Name	
Height	Height Above Ground	
Downtilt	Tilt	
EL Downtilt	none	0
Frequency	Frequency	
Power	Cell Power	
Misc. loss	none	0
Bandwidth	Bandwidth	
Noise figure	none	6
Downlink duplex factor	none	0.6
Subcarrier spacing	none	30
TX MIMO	TxMIMO	
RX MIMO	RxMIMO	
Active antenna effect	none	6
Cell load	none	30

Color index	none	
Technology	Tech	
Prediction model ID	none	3
Prediction model configuration ID	none	4
Frequency group	frequency_group	
Antenna ID	none	515
Carriers	none	[]
Site ID	none	
Duplex mode	Duplex	
Status	none	
Type	none	

4.2.6 Saving a Mapping Template

Before importing, save the mapping configuration:

1. In **Mapping presets**, click **+ New mapping preset**.



2. Enter the template name:

Import template

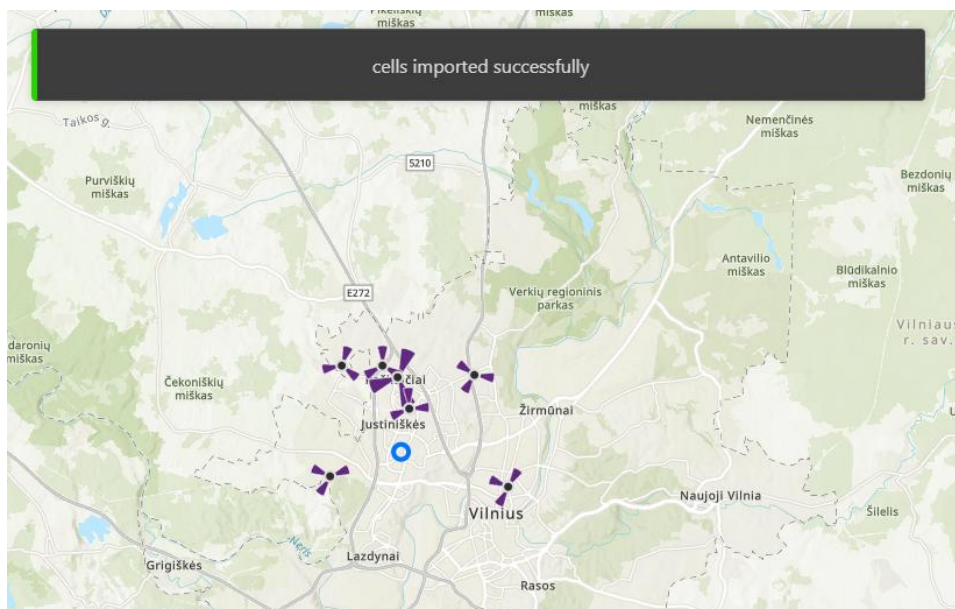


3. Click **Apply** to save the preset.

Using mapping templates ensures consistency and avoids repeating manual configuration for future imports.

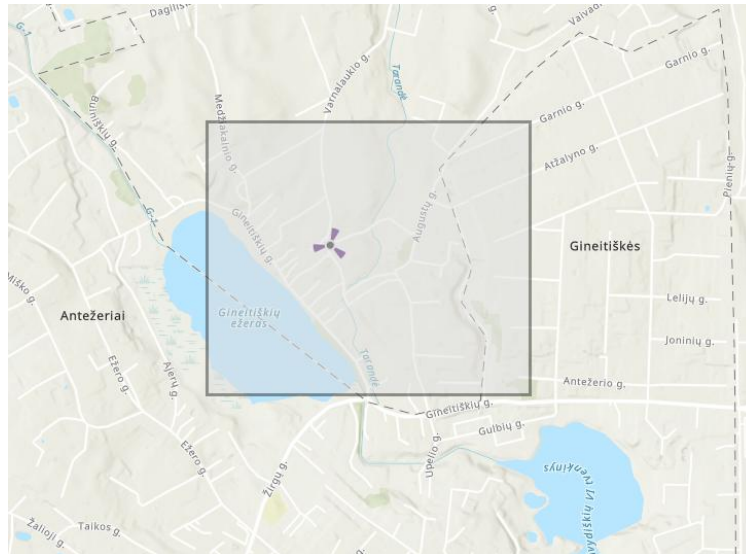
4.2.7 Importing the Data

1. Click **Accept** to start the import.
2. Wait until the import process completes.
3. Close the Import tool.

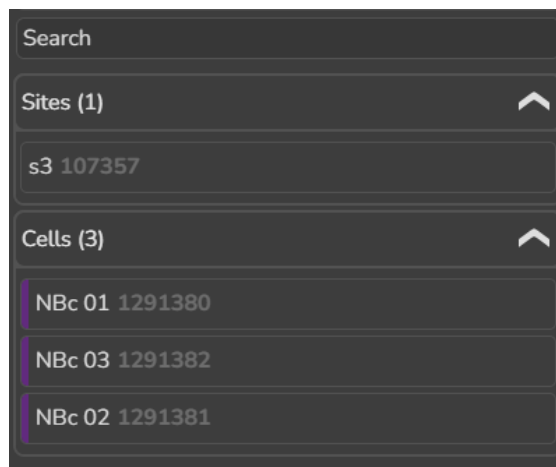


4.2.8 Verifying Imported Objects

1. Ensure the **Features** tool is active.
2. Click once on the map to select imported objects.



3. Review the objects listed in the Features panel.



This confirms that cells and sites were successfully created and loaded into the workspace.

4.3 Step 3 – Importing Antenna Patterns

4.3.1 Purpose of Antenna Import

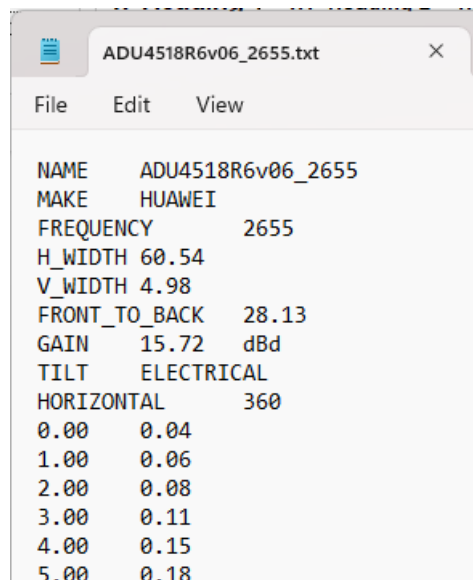
Antenna patterns define how energy is distributed spatially. Importing accurate antenna patterns ensures that predictions and analyses reflect real equipment behavior.

4.3.2 Reviewing the Antenna File

1. Navigate to:

C:\CE_Course\ImportingData\Antenna

2. Open the file **ADU4518R6v06_2655.txt** using Notepad.



The file is in **Planet format** and contains:

- Header information
- Horizontal radiation pattern
- Vertical radiation pattern

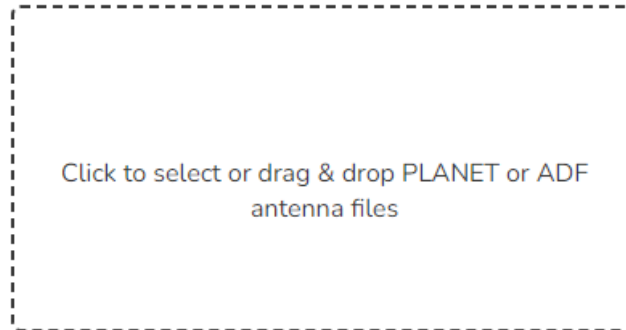
This format is supported for direct import into CE Express.

4.3.3 Importing the Antenna File

1. Open the **Antennas** tool.



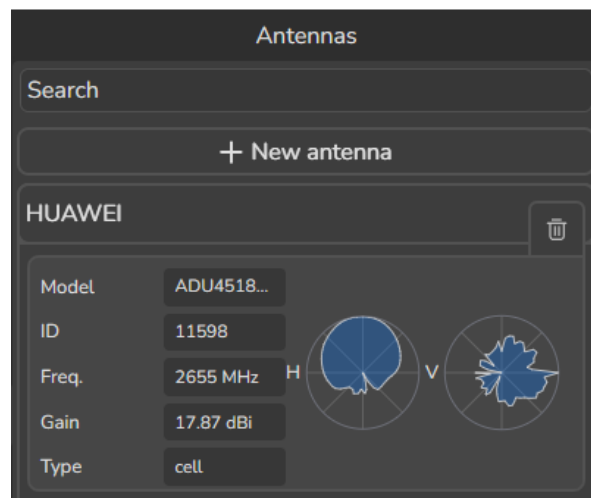
2. Click **+ New Antenna**.
3. Click **Import files**.
4. Drag and drop the antenna text file into the import area.



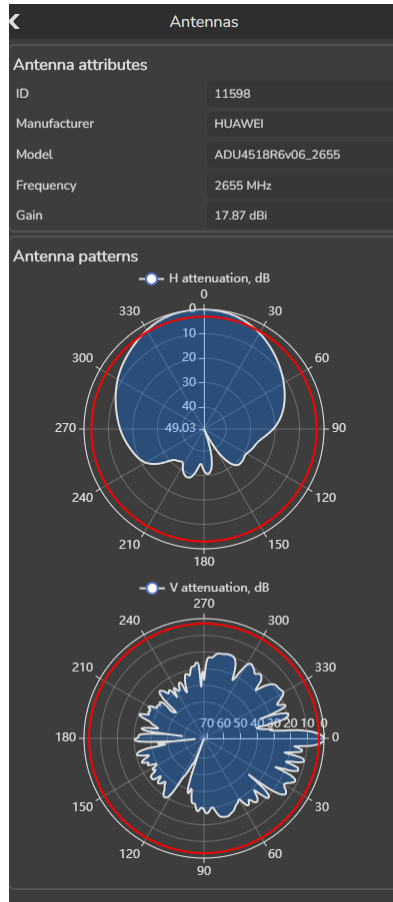
The antenna is automatically added to the CE Express antenna database.

4.4 Reviewing Imported Antenna Data

1. Select the imported antenna from the list.
2. Review antenna properties and radiation patterns.



Verifying antenna data ensures it is ready for assignment to cells and use in predictions.



3. Close the Antennas tool.

5. Summary and Key Takeaways

- Data import enables rapid population of workspaces using external datasets
- CSV import with explicit mapping ensures correct parameter assignment
- Mapping templates provide repeatability and consistency
- Antenna pattern import allows use of real equipment characteristics
- Verifying imported data is essential before running predictions or further analysis